N0JCG AIR TRAFFIC CENTER

END USER GUIDE

Operate the map, traffic sources, weather radar, and receive-only aviation audio

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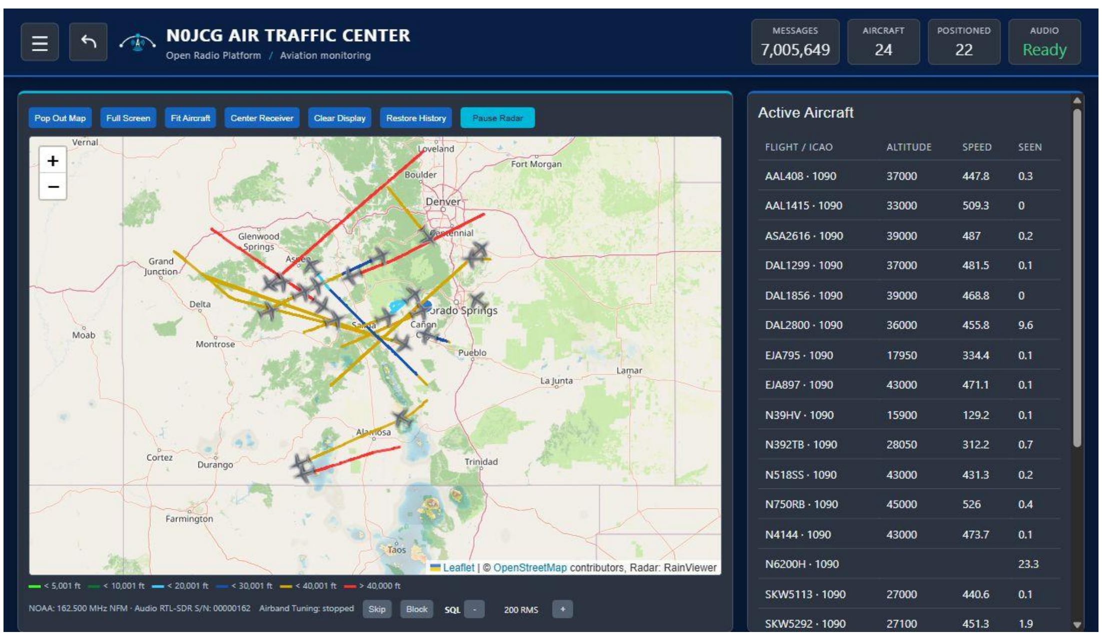


Live dashboard overview captured from the ROC test deployment | N0JCG Air Traffic Center v1.1.1

SCOPE This guide is for normal operation of the browser interface. Pi installation, receiver ownership, and recovery procedures are included for maintainers; browser screenshots show UI state only and do not by themselves prove antenna, decoder, or audio hardware health.

At a glance

N0JCG Air Traffic Center is a Raspberry Pi 5 browser application for receive-only aviation monitoring. It combines aircraft traffic from ADS-B and optional UAT, a weather radar overlay, NOAA Weather Radio and civil Airband controls, and a detail view for aircraft enrichment.

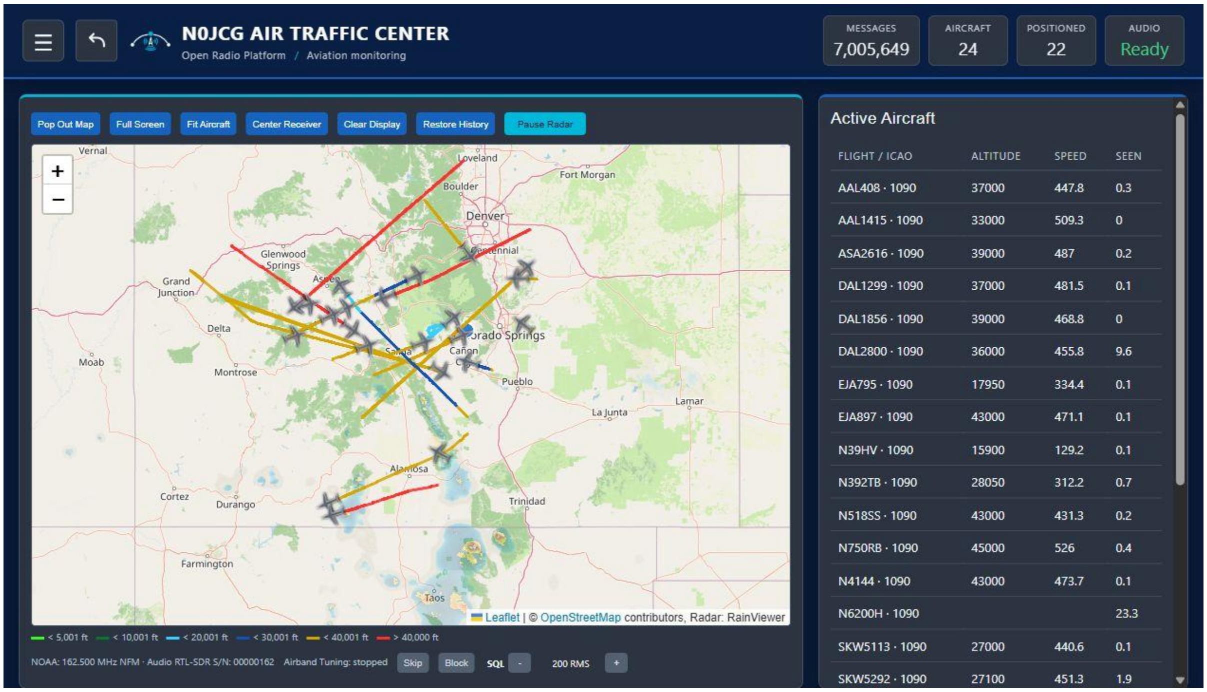
Item	Value
Default browser URL	http://<pi-lan-ip>:8090
Service	pi-air-traffic-tracker.service
Product version	1.1.1
Primary platform	Raspberry Pi 5 / Linux
Operating boundary	Receive-only monitoring; not a navigation or safety-of-flight system

Contents

QUICK START Open the browser URL, confirm the header status reads Audio Ready when audio is expected, check the traffic counters, and select an aircraft row to inspect its details. Use the menu for receiver controls and the ROC back button to return to the parent dashboard.

1. Understand the dashboard

The dashboard is state-first: the header and status cards expose what the app believes is active, while the map and table provide the operational view. A live deployment can have no aircraft, incomplete enrichment, or unavailable audio while the web page itself remains reachable.



Dashboard: branded header, status cards, map, aircraft trails, and active-aircraft list | N0JCG Air Traffic Center v1.1.1

Area	What to look for
Header	N0JCG identity, menu control, and the back button to the ROC dashboard.
Status cards	Messages, aircraft count, positioned count, and audio readiness.
Map	Aircraft markers, trails, receiver location, and optional radar overlay.
Active aircraft	Rows can be selected to open a detail card; values depend on decoded and enriched data.

EXPECTED RESULT The page loads without a blank shell, the map has a usable base layer, and the status cards update as the backend publishes data.

2. Getting started with a new Pi

Use this section when preparing a new standalone N0JCG Air Traffic Center installation. The guided installer handles the software sequence, but the operator must prepare the Pi, connect the correct receivers and antennas, and identify the FlyCatcher paths before serial programming.

Hardware checklist

Item	Recommendation
Raspberry Pi	Raspberry Pi 5 with a reliable USB-C power supply and adequate cooling.
SD card	High-endurance 32 GB or larger card; use a quality card reader and verify the image before first boot.
Network	Ethernet is recommended for the initial setup and steady aircraft/audio operation. Wi-Fi can be used when Ethernet is unavailable.

ADS-B / UAT receiver	Nooelec FlyCatcher dual-tuner receiver with clearly identified ADS-B and UAT paths.
NOAA receiver	Nooelec NESDR Nano2+ assigned to 00000162 for NOAA Weather Radio.
Airband receiver	Dedicated RTL receiver assigned to 00000118 for civil Airband.
USB accessories	Powered USB hub if needed; keep the Pi power supply and receiver cabling physically secure.

Recommended antennas

- 1090 MHz: a dedicated outdoor or window-mounted ADS-B antenna with low-loss coax and a clear view of the sky.
- 978 MHz: a band-appropriate antenna or the FlyCatcher antenna path intended for UAT reception.
- 162 MHz NOAA: a VHF antenna covering approximately 162 MHz, connected to the NESDR Nano2+.
- Civil Airband: a VHF antenna covering approximately 118-137 MHz, connected to the dedicated Airband receiver.
- Keep antenna cables short where practical and label each cable with its receiver role before connecting USB devices.

RECEIVE-ONLY BOUNDARY This application only receives and displays aircraft and radio information. It does not transmit. Antenna placement, filtering, geography, and local RF conditions determine coverage and audio quality.

Prepare the Pi operating system

1. Use Raspberry Pi Imager to write the current 64-bit Raspberry Pi OS/Debian-family image to the SD card. Choose a Lite image for a headless appliance or a Desktop image if the Pi will have a local display.
2. In the Imager customization screen, set the hostname, create the Pi user, configure the network, set the time zone, and enable SSH using password authentication for the initial deployment.
3. Insert the SD card, connect Ethernet and power, and allow the Pi to complete its first boot.
4. From the workstation, confirm the Pi is reachable over SSH and keep all RTL-SDR receivers disconnected until the installer requests them.

Run the guided first deployment

Run this from the MSYS2/UCRT64 terminal on the workstation containing the N0JCG Air Traffic Center repository:

```
./tools/install_pi_initial_deployment.sh
```

1. Enter the Pi IP address, SSH user (normally pi), and SSH password. The launcher stores reusable values in a private local .env file.
2. Follow the prompts to insert only the named radio. The installer backs up its EEPROM, assigns the correct serial automatically, asks you to remove it, and verifies it once before continuing.
3. When prompted, reconnect all four receivers with their matching antennas. The installer validates the serial-to-role map before enabling the services.
4. After service and API validation pass, open http://<pi-ip-address>:8090 in a browser on the same LAN.

REQUIRED SERIAL MAP ADS-B 1090 uses 00001090; NOAA uses 00000162; dedicated Airband uses 00000118; UAT 978 uses 00000978. The installer uses stable EEPROM serials rather than transient Linux USB indexes.

3. Before you begin

The normal operator only needs a browser on the same LAN as the Pi. Maintainers working on the Pi should confirm the receiver labels and serials before starting or changing services.

Role	Receiver / source	Permanent serial
NOAA Weather Radio	Nooelec NESDR Nano2+; 162 MHz NFM	00000162
Civil Airband	Dedicated RTL receiver; civil Airband AM	00000118
ADS-B 1090	Nooelec FlyCatcher ADS-B side; application-owned readsb	00001090
UAT 978	Nooelec FlyCatcher UAT side; dump978-fa when enabled	00000978

SAFETY BOUNDARY Do not swap serial assignments casually. Stop receiver-owning services before EEPROM work, back up each EEPROM, fully power-cycle after writes, and never run a second readsb service beside the application-owned decoder.

Receiver ownership

- The NOAA VHF antenna belongs on the Nano2+ assigned 00000162.
- The civil Airband antenna belongs on the dedicated receiver assigned 00000118.
- The 1090 MHz antenna belongs on the FlyCatcher ADS-B side assigned 00001090.
- The 978 MHz antenna belongs on the FlyCatcher UAT side assigned 00000978.
- NOAA and Airband have dedicated receivers: 00000162 and 00000118.

4. Open and validate the app

Open the browser interface

Use the current LAN address of the Pi. The default port is 8090.

```
http://<pi-lan-ip>:8090
```

For a known installation, the URL may look like:

```
http://192.168.68.110:8090
```

Validate the Pi service

These checks are for a maintainer with shell access. Run them on the Pi, not in the browser address bar.

```
sudo systemctl status pi-air-traffic-tracker.service --no-pager
curl -fsS http://127.0.0.1:8090/api/status | jq
```

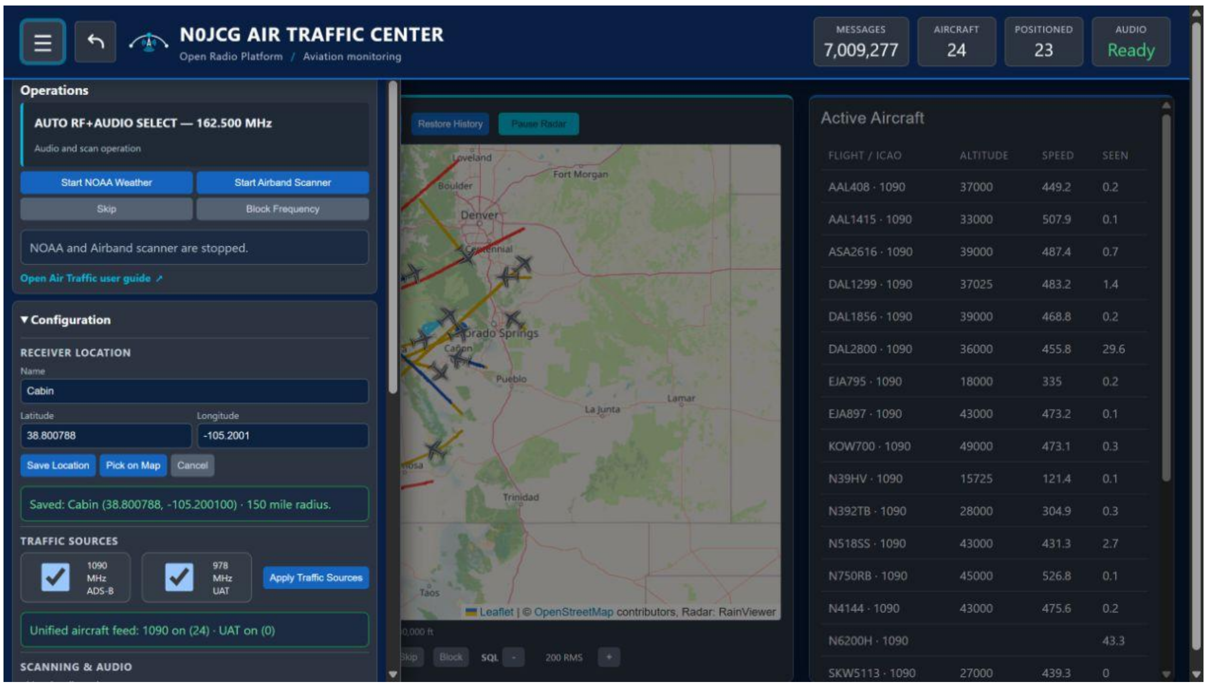
- Confirm the service is active and the API returns JSON.
- Confirm receiver_roles identify 00000162, 00000118, 00001090, and 00000978 as configured.
- If the service is active but the UI is empty, inspect the service journal and receiver ownership before changing browser settings.

RECOVERY A browser refresh is safe for a stale page. A service restart is an administrative action and can interrupt active receiver and audio sessions; use it only when the service or backend requires recovery.

5. Operate traffic and audio sources

Use the operator menu

Select the menu button in the header. The drawer groups operational controls below the app identity and keeps the map visible behind a dimmed backdrop.



Operator menu: navigation, user guide link, audio controls, receiver location, and traffic sources | N0JCG Air Traffic Center v1.1.1

The menu includes the user guide link and the back button target used to return to the ROC dashboard. Close the drawer with the menu button or its normal close action.

NOAA Weather Radio

1. Open the NOAA Weather controls from the menu.
2. Select Start NOAA Weather and choose a scan or frequency as provided by the deployment.
3. Confirm the UI reports an active state before expecting audio.
4. Stop any active NOAA listening session before starting civil Airband; Airband uses dedicated receiver 00000118.

Civil Airband

1. Stop NOAA if it is active.
2. Open Airband controls and start the receive-only audio mode.
3. Use squelch and skip/block controls to manage the listening experience; the exact controls available depend on the current UI state.

EXPECTED RESULT Audio readiness is explicit in the header/status area. An enabled control is not the same as confirmed RF reception; verify with the active state, service logs, and the connected audio path when troubleshooting.

6. Registration and trial access

Unregistered installations include a manually restarted five-minute evaluation period. The trial controls aircraft tracking, NOAA Weather Radio, civil Airband scanning, and shared audio. When the timer expires, the backend stops those functions and the browser clears the aircraft list, map markers, and trails.

Register the installation

1. Open the menu and expand Registration.
2. Enter the license S/N and the registered email address. The application trims the serial, converts it to uppercase, and normalizes the email before activation.
3. Select Activate license and wait for the registration status to confirm success.
4. After successful activation, the Restart Trial control is hidden and the licensed installation can continue operating without the trial timer.

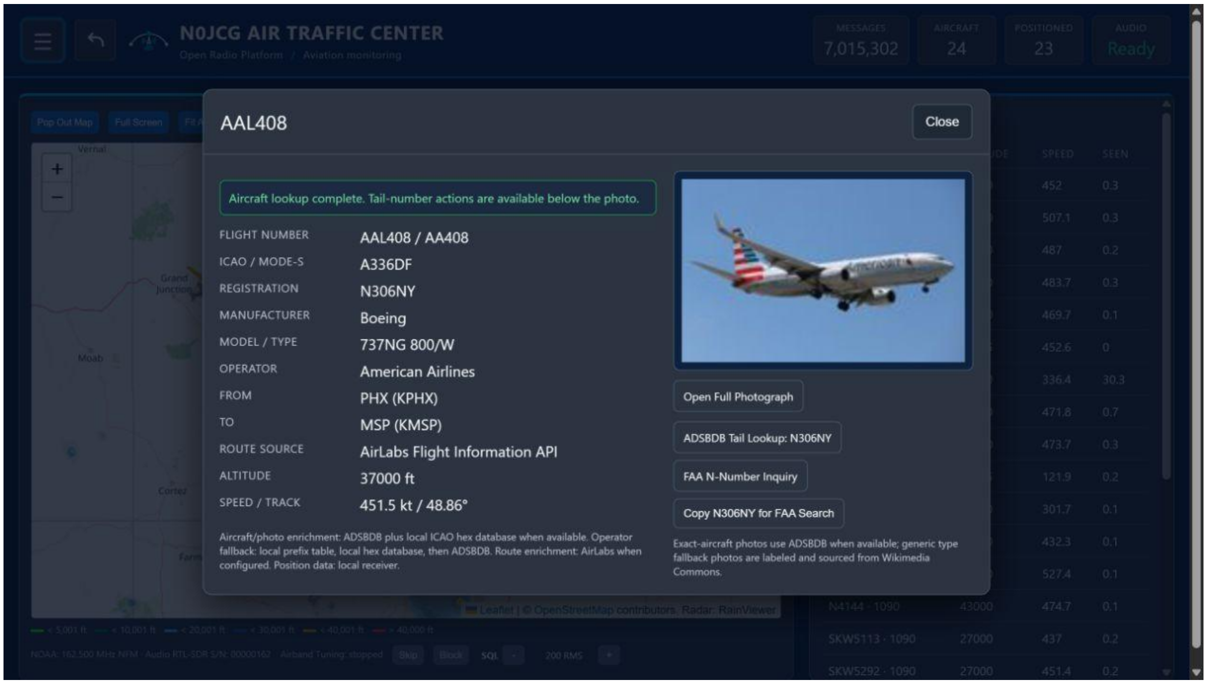
PRIVACY AND VALIDATION The browser submits only the license S/N and email to the local Pi endpoint. The backend adds the installation identity, product slug, and application version before contacting the N0JCG licensing service and verifies the signed lease before accepting it.

Restart the free trial

- For an unregistered installation, select Restart Trial in the header to begin another five-minute period manually.
- The button is intentionally not an automatic renewal and is unavailable after registration.
- If the trial expires, register the product or restart the trial before expecting aircraft or audio activity to resume.

7. Use the aircraft detail view

Select an aircraft marker or active-aircraft row to open its detail card. The card presents the data currently available from decoded traffic and optional enrichment services.



Aircraft detail card: identity, route, position, and available enrichment | N0JCG Air Traffic Center v1.1.1

Field group	Meaning
Identity	Flight, ICAO address, registration, type, and operator when available.
Route	Departure and destination codes when enrichment has a matching route.
Position	Altitude, speed, track, and last-update information from decoded traffic.
Photo / enrichment	Optional external data; availability and accuracy depend on the configured service.

INTERPRETATION Missing route, photo, or registration data is not automatically a decoder failure. Treat decoded position and enrichment as separate data paths and use the live status/API when diagnosing gaps.

8. Weather radar and map controls

Weather radar is a browser-side overlay and requires internet access from the browser. The application keeps the displayed frame visible while the next frame loads, so a slow network should not create a distracting blank or fade between frames.

- Use the Weather Radar menu to enable the overlay and adjust opacity.
- Use the map controls to pan and zoom without changing receiver configuration.
- Aircraft trails and radar are separate layers; disabling one should not remove the other.
- If radar is unavailable while the map works, check browser internet access and external tile/radar availability before restarting the Pi service.

EXPECTED RESULT The last usable radar frame remains visible during a normal frame transition. A browser or external tile failure may still prevent new frames from loading.

9. Maintenance and recovery

Routine service commands

```
sudo systemctl restart pi-air-traffic-tracker.service
sudo systemctl stop pi-air-traffic-tracker.service
sudo systemctl start pi-air-traffic-tracker.service
sudo journalctl -u pi-air-traffic-tracker.service -n 200 --no-pager
```

Common symptoms

Symptom	First checks	Recovery
Receiver busy	Check competing readsb/dump1090 services and USB ownership.	Stop the conflicting service; keep the app-owned decoder as the owner.
No aircraft	Check receiver_roles, antenna connection, decoder logs, and recent messages.	Correct ownership/configuration, then restart the app service if needed.
No NOAA scan	Confirm NOAA VHF antenna and serial 00000162; inspect USB claim errors.	Stop other audio mode and retry after service health is confirmed.
Radar blank	Check browser internet and radar/tile availability.	Refresh or toggle the overlay.

Update workflow

1. Back up local configuration and note the current version.
2. Deploy the intended standalone repo revision to the Pi.
3. Run the project installer/validator and confirm the service starts.
4. Open the browser URL and compare status, map, traffic list, and menu behavior.
5. Record any receiver or audio validation separately from browser UI validation.

10. Operator checklist and limitations

Use this short checklist after a deployment, reboot, or significant configuration change.

- The browser opens at the current Pi LAN address on port 8090.
- The N0JCG header, menu, ROC back button, and user guide link are visible.
- Status cards update and the API is reachable on the Pi.
- ADS-B 1090 is owned by serial 00001090; UAT 978 is 00000978 when enabled.
- NOAA owns 00000162 and civil Airband owns dedicated receiver 00000118.
- Aircraft rows open detail cards without confusing missing enrichment for missing RF data.
- Radar behavior is checked from the browser network path, not inferred from receiver state.

LIMITATIONS N0JCG Air Traffic Center is a receive-only monitoring tool. It is not a certified navigation, separation, emergency, weather-warning, or safety-of-flight system. Do not use its display as a substitute for official aviation information or operational procedures.

Reference paths

Resource	Location
Source repository	N0JCG-AIR-TRAFFIC-CENTER (legacy checkout: PI-AIR-TRAFFIC-TRACKER)
Service	pi-air-traffic-tracker.service
API status	http://127.0.0.1:8090/api/status on the Pi
Detailed maintainer guide	docs/PI_AIR_TRAFFIC_TRACKER_USER_GUIDE.md
Release scope	docs/RELEASE_V1_1_1.md

Screenshot note: the screenshots in this publication were captured from the ROC test deployment on August 10, 2026. Live traffic values, aircraft identities, radar frames, and enrichment results will change over time.